

Research Publications: Avinash Tiwari

Short author works

- 1 **A. Tiwari**, S. A. Bhat, T. R. Choudhury, S. Adhikari, M. K. Singh, and S. J. Kapadia, “Gravitational Wave Informed Inference of 21-cm Global Signal Parameters,” Feb. 2026.  DOI: 10.1093/mnras/stag923 arXiv: 2602.07631 [astro-ph.CO].
- 2 **A. Tiwari** and S. J. Kapadia, “Coalescence Forensics: Weighing the Hosts of Hierarchical Binary Black Hole Mergers,” Feb. 2026, Under review. arXiv: 2602.05645 [astro-ph.HE].
- 3 **A. Tiwari**, S. J. Kapadia, A. Vijaykumar, and S. Chatterjee, “Periodic line-of-sight velocity-driven modulations to gravitational waves emitted by compact binaries in Keplerian outer orbits,” Jul. 2026. arXiv: 2607.09644 [gr-qc].
- 4 **A. Tiwari**, A. Vijaykumar, S. J. Kapadia, and S. Chatterjee, “Identifying and characterizing extragalactic circum-CBC exoplanets with future gravitational-wave detectors,” Jul. 2026. arXiv: 2607.09658 [astro-ph.HE].
- 5 M. A. Ansari, **A. Tiwari**, S. A. Bhat, S. Nair, and S. J. Kapadia, “Line-of-sight Acceleration Driven Effects in Eccentric Compact Binary Coalescences,” 2026, Under PnP Review (LIGO document number: P2600511).
- 6 M. A. Ansari, **A. Tiwari**, S. A. Bhat, A. Vijaykumar, and S. J. Kapadia, “Line-of-Sight Acceleration-driven waveform corrections for unequal-mass compact binary coalescences,” 2026, Under PnP Review (LIGO document number: P2600512).
- 7 A. Tiwari, A. Vijaykumar, S. J. Kapadia, S. Ghosh, and A. B. Nielsen, “Pipeline to search for signatures of line-of-sight acceleration in gravitational wave signals produced by compact binary coalescences,” *Phys. Rev. D*, vol. 113, no. 10, p. 104 072, 2026.  DOI: 10.1103/qtw8-3g56 arXiv: 2506.22272 [astro-ph.HE].
- 8 **A. Tiwari**, S. A. Bhat, M. A. Shaikh, and S. J. Kapadia, “Testing the Nature of GW200105 by Probing the Frequency Evolution of Eccentricity,” *Astrophys. J.*, vol. 995, no. 1, p. 48, 2025.  DOI: 10.3847/1538-4357/ae1d74 arXiv: 2509.26152 [astro-ph.HE].
- 9 **A. Tiwari**, P. Chanda, S. J. Kapadia, S. Adhikari, A. Vijaykumar, and B. Dasgupta, “Profiling Dark Matter Spikes with Gravitational Waves from Accelerated Binaries,” Aug. 2025, Under review. arXiv: 2508.03803 [hep-ph].
- 10 **A. Tiwari**, A. Vijaykumar, S. J. Kapadia, S. Chatterjee, and G. Fragione, “Profiling stellar environments of gravitational wave sources,” *Physical Review D*, vol. 112, no. 8, Oct. 2025, ISSN: 2470-0029.  DOI: 10.1103/gspl-m478
- 11 S. A. Bhat, **A. Tiwari**, M. A. Shaikh, and S. J. Kapadia, “Eccentricity evolution consistency test to distinguish eccentric gravitational-wave signals from eccentricity mimickers,” *Phys. Rev. D*, vol. 112, no. 12, p. 124 004, 2025.  DOI: 10.1103/rstg-6n6y arXiv: 2508.14850 [gr-qc].
- 12 **A. Tiwari**, A. Vijaykumar, S. J. Kapadia, G. Fragione, and S. Chatterjee, “Accelerated binary black holes in globular clusters: forecasts and detectability in the era of space-based gravitational-wave detectors,” *Mon. Not. Roy. Astron. Soc.*, vol. 527, no. 3, pp. 8586–8597, 2023.  DOI: 10.1093/mnras/stad3749 arXiv: 2307.00930 [astro-ph.HE].
- 13 A. Vijaykumar, **A. Tiwari**, S. J. Kapadia, K. G. Arun, and P. Ajith, “Waltzing Binaries: Probing the Line-of-sight Acceleration of Merging Compact Objects with Gravitational Waves,” *Astrophys. J.*, vol. 954, no. 1, p. 105, 2023.  DOI: 10.3847/1538-4357/acd77d arXiv: 2302.09651 [astro-ph.HE].